Transpose duality for preorder multichain matrices

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Abstract

We prove the matrix Ehrhart–zeta duality conjecture of Athanasiadis and Chapoton. Coordinate increments identify multichains with lattice points in a sum of simplices attached to a bipartite graph with repeated vertices. Exchanging the two vertex classes reverses the preorder and exchanges the two parameters. Postnikov's established lattice-point duality then gives the claimed transpose.

Task 1. Full solution

Let τ be a preorder on a finite set E. Its preorder polytope is

$$\mathcal{Q}_\tau = \{x \in \mathbb{R}^E_{\geq 0} : \sum_{e \in I} x_e \leq |I| \text{ for every order ideal } I \text{ of } \tau\}.$$

For $k, \ell \geq 0$, let $\nabla_\tau(k, \ell)$ count sequences

$$a^{(1)} \leq a^{(2)} \leq \cdots \leq a^{(k)}, \qquad a^{(j)} \in \ell\mathcal{Q}_\tau \cap \mathbb{Z}^E,$$

where comparison is coordinatewise. These are the k-multichains of [1, Section 4.2]. Let τ^* be the reversed preorder.

Theorem 1. *For every finite preorder and all $k, \ell \geq 0$,*

$$\boxed{\nabla_\tau(k, \ell) = \nabla_{\tau^*}(\ell, k).}$$

Thus Conjecture 4.2 of [1] holds.

The established bipartite-graph duality

Let G have disjoint vertex classes A, B, and let $N(S)$ be the neighbors of a subset of either class in the opposite class. For a set $J \subseteq B$, put $\Delta_J = \mathrm{conv}(0, e_b : b \in J)$ and define

$$P_G = \sum_{a \in A} \Delta_{N(a)} \subseteq \mathbb{R}^B.$$

Let G^* be the graph with its two vertex classes exchanged. We use Postnikov's lattice-point duality

$$\#(P_G \cap \mathbb{Z}^B) = \#(P_{G^*} \cap \mathbb{Z}^A), \tag{1}$$

recorded explicitly in [2, Lemmas 2.4–2.6]. We recall the characterization of its integer points and derive (1) from the counting theorem used by the primary source.

First,

$$P_G \cap \mathbb{Z}^B = \left\{ z \in \mathbb{Z}^B_{\geq 0} : \sum_{b \in S} z_b \leq |N(S)| \text{ for every } S \subseteq B \right\}. \tag{2}$$

Necessity holds because a simplex summand contributes at most one to the sum over S, and contributes zero unless its index lies in $N(S)$. Conversely, replace each b by z_b copies. Every subset of copies projecting to S has at most $\sum_{b \in S} z_b \leq |N(S)|$ members. Hall's marriage theorem therefore matches all copies to distinct vertices of A. For each matched a, choose its matched vector e_b from $\Delta_{N(a)}$, and choose 0 for unmatched a. Their sum is z, proving (2).

To see (1) from [1, Theorem 2.1], lift the simplices to the coordinates $B \sqcup \{0\}$, using the sets $I_a = N(a) \sqcup \{0\}$ with coefficient 1, and adjoin the universal set $I_0 = B \sqcup \{0\}$ with coefficient 0. Projection forgetting coordinate 0 is a bijection of integer points, since the sum of all coordinates is fixed at $|A|$. The draconian vectors in that theorem are

$$d_0 + \sum_{a \in A} d_a = |B|, \qquad d_a \geq 0, \qquad \sum_{a \in T} d_a \leq |N(T)| \quad (T \subseteq A).$$

Indeed, the inequalities for nonempty $T \subseteq A$ follow after canceling the extra vertex 0; those involving the universal set follow from the total-sum condition. The inequality for $T = A$ ensures that d_0 is uniquely determined and nonnegative. At these coefficients, every binomial factor in Theorem 2.1 is 1. The number of lattice points is therefore the number of vectors $(d_a)_{a \in A}$ satisfying the displayed inequalities. By (2) for G^*, this is exactly the right side of (1). This graph duality is an established input, not a new claim of this solution.

The graph of repeated preorder elements

Assume $k, \ell \geq 1$. Form $G = G_\tau(k, \ell)$ with

$$A = E \times [\ell], \qquad B = E \times [k],$$

and join $(e, s) \in A$ to $(f, t) \in B$ exactly when $e \leq_\tau f$. For $S \subseteq B$, let $\pi(S) \subseteq E$ be its projection. If $D(T)$ denotes the order ideal generated by $T \subseteq E$, then

$$N(S) = D(\pi(S)) \times [\ell].$$

Consequently (2) reduces to

$$z_{f,t} \geq 0, \qquad \sum_{f \in I} \sum_{t=1}^{k} z_{f,t} \leq \ell |I| \quad (I \text{ an order ideal of } \tau). \tag{3}$$

Here is the reduction in both directions. Taking $S = I \times [k]$ in (2) gives (3), since $D(I) = I$. Conversely, for arbitrary S, take $I = D(\pi(S))$. Nonnegativity and $S \subseteq I \times [k]$ give

$$\sum_{(f,t) \in S} z_{f,t} \leq \sum_{f \in I, t \in [k]} z_{f,t} \leq \ell |I| = |N(S)|,$$

as required.

An integer vector in (3) determines the chain

$$a_f^{(j)} = \sum_{t=1}^{j} z_{f,t} \qquad (1 \le j \le k). \tag{4}$$

The terminal vector belongs to $\ell\mathcal{Q}_\tau$ by (3). Every earlier vector does too, since the polytope is closed under decreasing nonnegative coordinates. Conversely, a multichain gives the nonnegative increments $z_{f,t} = a_f^{(t)} - a_f^{(t-1)}$, where $a^{(0)} = 0$. These two constructions are inverse, so

$$\nabla_\tau(k,\ell) = \#(P_{G_\tau(k,\ell)} \cap \mathbb{Z}^{E\times[k]}). \tag{5}$$

Proof of Theorem 1. Exchanging the two graph classes gives exactly

$$G_\tau(k,\ell)^* = G_{\tau^*}(\ell,k),$$

because $e \le_\tau f$ is equivalent to $f \le_{\tau^*} e$. Apply (1) and then (5) to both graphs. This proves the theorem for positive parameters. If $k = 0$ there is one empty multichain; if $\ell = 0$, the lattice-point poset contains only 0 and again there is one multichain. These cases give the same identity. The empty ground set, if allowed, also gives the constant count 1. □

Scope, attribution, and verification

No antisymmetry assumption was used, so equivalent distinct elements of the preorder are included. The argument proves every matrix entry for arbitrary parameters. The conjecture and polytope definitions are due to Athanasiadis and Chapoton. Postnikov's counting theorem and Hall's theorem are established inputs. The repeated-vertex graph and increment identification are the reduction resolving the target.

All 26 pages of the current primary were read, along with the relevant graph-duality statements in [2]. Four exact verification groups pass: direct chain enumeration, 9,725 transpose identities, independent graph and increment-vector enumeration, and further graph examples. Bounded checks found no prior full target claim or identified journal final. Review is by the authoring agent only, without independent peer review or proof-assistant certification.

Tasks 2–5. Proposed follow-up directions

Task 2. Turn Postnikov's graph duality into an explicit algorithm mapping individual multichains between the two sides.

Task 3. Determine refinements that retain the ranks of the chain entries and remain compatible with exchanging the graph classes.

Task 4. Study versions with nonuniform numbers of copies of each preorder element, allowing different chain capacities by coordinate.

Task 5. Derive effective formulas for these matrices in families of preorders with small width or bounded graph complexity.

References

[1] C. A. Athanasiadis and F. Chapoton, *Polytopes and posets associated to preorders*, arXiv:2605.26916v1, 2026.

[2] Z. Dai, Q. Hou, Z. Liu, W. Thawinrak, and H. Wang, *Counting Lattice Points in Minkowski Sums of Cross Polytopes*, arXiv:2608.16037v2, 2026.